

Generation Interconnection System Impact Study Report

For Queue Project Q20-1432

Picnic Point 100 kV

May 21, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1432. The Interconnection Customer proposes a 63.8 MW battery energy storage facility interconnecting at the Picnic Point 100 kV substation in the Ohio area. The request was received on March 17, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 504 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,285,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Carbon Center 500-345 kV Autotransformer	Blackrock – Shawville Station 345.0 kV line	114.5	1,998	5.7
Shawville Sta Substation – Carbon Ceter 500.0 kV	Blackrock – Shawville Sta 345.0 kV	107.9	2,111	5.7
Orchard Rd – Shawville Sta Substation 500.0 kV line	Scott – Shawville Sta 345.0 kV	105.0	2,944	7.2
Picnic Pt – Boswell Dairy Ave 100.0 kV Cir. 1	Boswell Dairy Ave – Picnic Pt 100.0 kV Cir. 3	103.0	162	10.4
Boswell Dairy Ave – Picnic Pt 100.0 kV L	Boswell Dairy Ave – Picnic Pt 100.0 kV #3	103.0	162	10.4
Boswell Dairy Ave – Picnic Pt 100.0 kV	Boswell Dairy Ave – Picnic Point 100.0 kV line	103.0	162	10.4
W Siloam Springs – Bogue 100.0 kV	West Siloam Springs – Jarales 100.0 kV L	102.9	142	8.2
Hortonville – Picnic Pt 100.0 kV L	Brincken'S Corner – Narcoossee 100.0 kV Line	102.3	146	10.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Carbon Ctr 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$33,680,000	0% (\$182,000)
Shawville Sta Substation – Carbon Ceter 500.0 kV	Reconductor line	\$33,337,000	0% (\$180,000)
Orchard Road – Shawville Sta 500.0 kV Line	Reconductor line	\$83,446,000	0% (\$444,000)
Boswell Dairy Ave – Picnic Pt 100.0 kV Line 1	Reconductor line	\$487,000	14% (\$68,000)
Boswell Dairy Ave – Picnic Pt 100.0 kV L	Reconductor line	\$985,000	14% (\$137,000)
Boswell Dairy Ave – Picnic Pt 100.0 kV	Reconductor line	\$643,000	14% (\$89,000)
W Siloam Springs – Bogue 100.0 kV	Reconductor line	\$508,000	11% (\$57,000)
Hortonville – Picnic Pt 100.0 kV L	Reconductor line	\$1,085,000	12% (\$128,000)
	Total allocated to project		\$1,285,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.